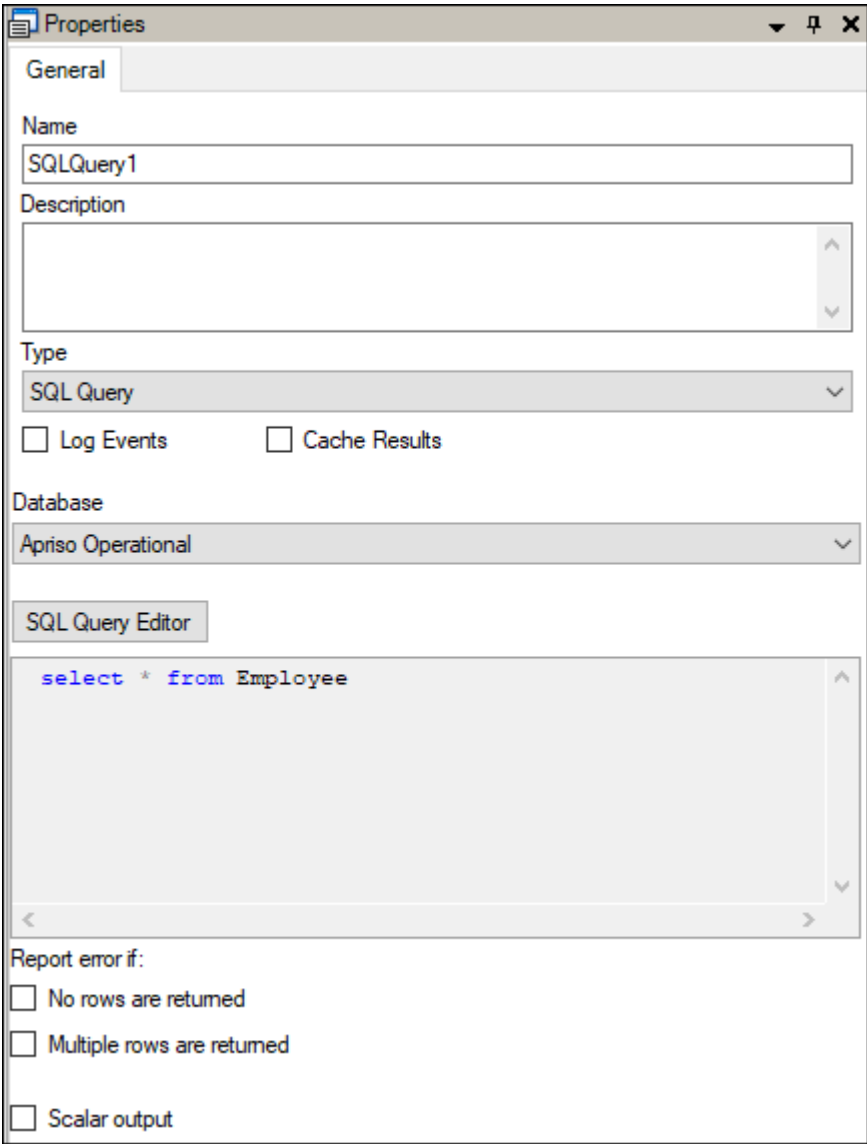


SQL Query Function

The SQL Query Function enables executing a user-defined query against the database and returning results. Each column of the query is assigned to a separate Function Output.

To Add the SQL Query Function

- 1. Drag the SQL Query Function from the [Toolbox](#) to the Function Navigation diagram, or double-click the SQL Query icon in the Toolbox.
- 2. Define the query in the SQL Query Editor. The Inputs and Outputs are automatically added in the Editor based on the query.
- 3. Choose the error reporting options for the Function. Optionally, choose the database alias to be used in runtime.

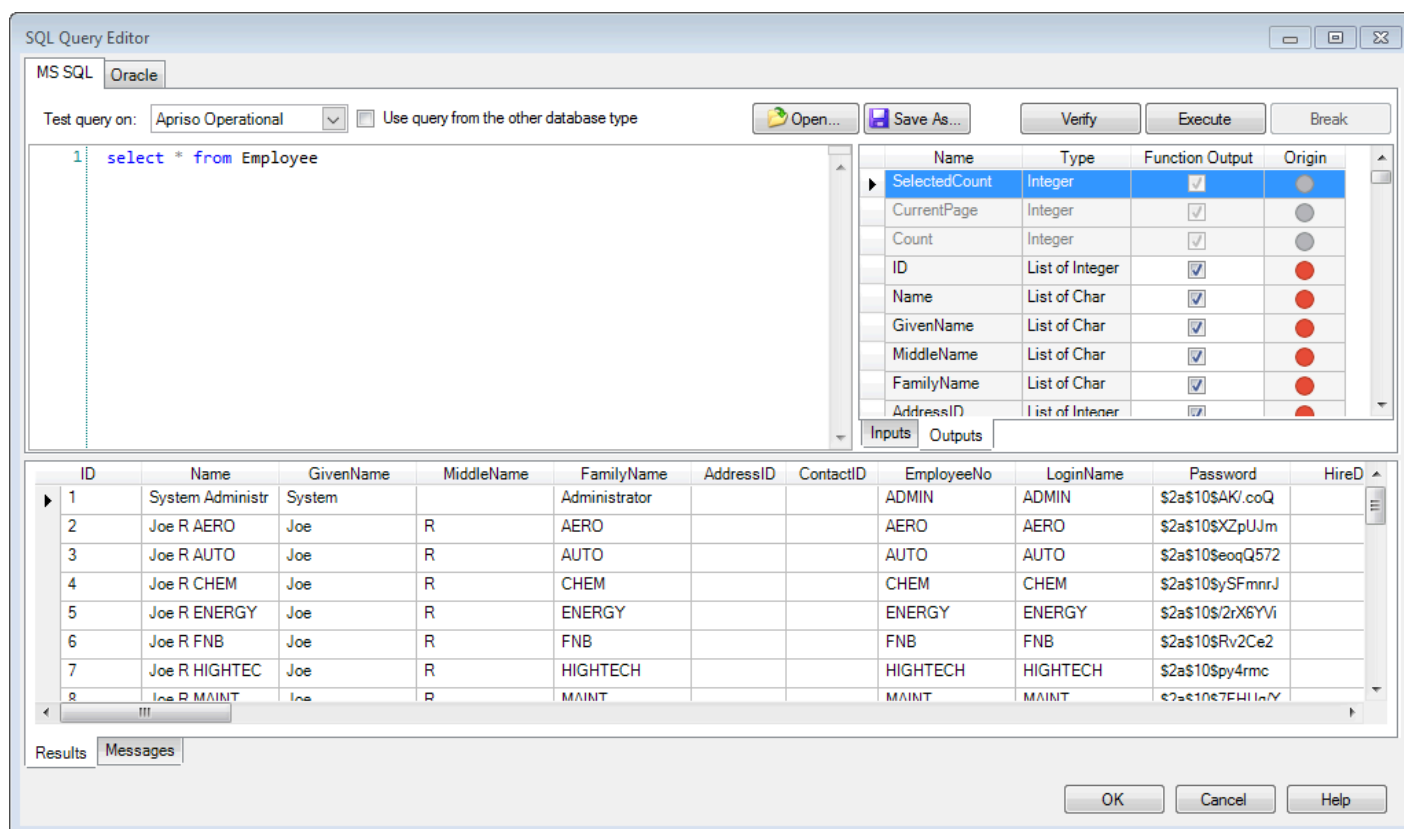


Field	Description
Database	The list of available database aliases defined in the "DataServices" section of the Central Configuration file (for details see Central Configuration Documentation). You can choose the database against which the query will be executed.
SQL Query field	The query defined for the current database type.

No rows are returned	If the query returns no rows and this check box is selected, the Output <code>IsError</code> returns True and the Output <code>ErrorCode</code> returns a proper code (NO_ROWS).
Multiple rows are returned	If the query returns multiple rows and this check box is selected, the Output <code>IsError</code> returns True and the Output <code>ErrorCode</code> returns a proper code (MULTIPLE_ROWS).
Scalar output	The user can configure the Function to always return scalar values (one row output). When this option is used, the two check boxes above are automatically selected. When the option of scalar values is selected, the system changes the Function Outputs types to non-list in both the properties of the Function and the SQL Query Editor.

SQL Query Editor

In the SQL Query Editor, you can write, verify, and execute SQL queries.



Field	Description
MS SQL / Oracle tab	You can write <code>sql_query_rich_editor</code> different queries for MS SQL and Oracle databases.
Test query on drop-down list	The database connection alias to be used for testing the query. You can add other databases to the list using the keys located in the "DataServices" section of the Central Configuration file (for more information and predefined aliases, refer to the Central Configuration Documentation).
Use a query from the other database type check box	If this box is checked, the system will use the query defined for the other database type (in the other tab). The box is checked by default in the Oracle tab.
Open button	Opens the query previously saved by the user.
Save as button	Saves the query written by the user.
Verify button	Checks if the query is valid and can be executed. The keyboard shortcut is <code>CTRL+F5</code> .

Execute button	Executes the query. The keyboard shortcut is F5 .
Break button	Cancels the pending execution of the query. The keyboard shortcut is ALT+PAUSE/BREAK .
Query body	This is where you can write your query. Elements of the query syntax will be highlighted. Standard Windows functions can be used, such as Copy, Paste, Cut, Delete, Select All, and Undo from both the right-click menu and the keyboard. It is also possible to find expressions in the formula using the CTRL+F Keyboard Shortcut. The following rules and restrictions apply: ▶ Only retrieving (filtering) data statements (e.g., SELECT) can be used ▶ Input names can be used in the query in two ways: by preceding the Input name with the "@" character (for variable binding), or by placing it in braces "{ }" (for text concatenating). Be careful when building a query with the use of {[your parameter]}. Remember that queries are executed directly on the database, and incorrect use of this functionality can result in accessing unwanted Apriso tables ▶ Multi-line comments (/*...*/) are not supported and can cause errors in runtime. It is highly recommended to remove them or replace them with line comments (--) Queries that return too many records or too long string in a cell might not work correctly in the SQL Query Editor. This is a third-party issue.
Inputs/Outputs tab	The list of Inputs and Outputs used in the SQL Query Function generated automatically based on the query. You can check the box next to individual column names to add them as Outputs in the Function.
Results tab	The result of the query.
Messages tab	The list of validation and error messages.

SQL Query Function Outputs

Input/Output	Description
Count	Contains the number of rows returned by the query.
IsError	Returns True if the error reporting for the Function is turned on and the error condition is fulfilled. This Output can be used directly as an Input to the Show Message Function .
ErrorCode	Upon an error, returns one of the following values depending on the query's result: ▶ NO_ROWS ▶ MULTIPLE_ROWS This Output can be used directly as an Input to the Show Message Function.